Project - 5

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★ Theory

1. Student Score as a vector.

Example $v_1 = [78, 82, 75, 80, 85, 90]$

$v_2 = [65, 70, 68, 72, 74, 78]$

$\vdots$

$v_{100} = [58, 62, 60, 64, 67, 69]$

2. Norm 1 & Norm 2

⇒ L1 Norm [Manhattan]

$$||v_1||_1 = 78 + 82 + 75 + 80 + 85 + 90$$
$$= 490$$

⇒ L_2 Norm (Euclidean)

$$||v_2||_2 = \sqrt{(78)^2 + (82)^2 + (75)^2 + (80)^2 + (85)^2 + (90)^2}$$
$$= \sqrt{40158}$$
$$= 200.39$$

3. Dot Product

$$v_1 \cdot v_2 = (78 \times 65) + (82 \times 70) + (75 \times 68) + (80 \times 72) + (85 \times 74) + (90 \times 78)$$

$$= 34980$$

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4. Angle between vectors:

$$\cos\theta = \frac{v_1 \cdot v_2}{\|v_1\| \, \|v_2\|}$$

$$\cos\theta = \frac{34980}{(200.39)(175.74)}$$

$$\cos\theta = 0.994$$

$$\theta \approx 6^\circ$$

5. Cross Product

$v_1 = [78, 82, 75]$ $\quad v_2 = [65, 70, 68]$

$$v_1 \times v_2 = \begin{vmatrix} i & j & k \\ 78 & 82 & 75 \\ 65 & 70 & 68 \end{vmatrix}$$

$$= (82 \times 68 - 75 \times 70)i - (78 \times 68 - 75 \times 65)j + (78 \times 70 - 82 \times 65)k$$

$$= 326i - 429j + 130k$$

6. Projection

$$\text{Proj}_{v_2}(v_1) = \frac{v_1 \cdot v_2}{|v_2|^2} \times \overrightarrow{v_2}$$

$$= \frac{34980}{30884.5476} \times [65, 70, 68, 72, 74, 78]$$

$$= 1.1326 \times [65, 70, 68, 72, 74, 78]$$

$$= [74.56, 80.30, 78, 82.59, 84.89, 89.48]$$

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★ Matrix Operations

$$A = \begin{bmatrix} 78 & 82 & 75 & 80 & 85 & 90 \\ 65 & 70 & 68 & 72 & 74 & 78 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

a) Addition

eg. $\begin{bmatrix} 78+65 & 82+70 & 75+68 & 80+72 & 85+74 & 90+78 \end{bmatrix}$

b) Multiplication

- $A^T . A \Rightarrow 6 \times 6$ matrix

c) Transpose

$$\begin{bmatrix} 78 & 65 & \cdots \\ 82 & 70 & \cdots \\ 75 & 68 & \cdots \\ 80 & 72 & \cdots \\ 85 & 74 & \cdots \\ 90 & 68 & \cdots \end{bmatrix}_{6 \times 100}$$

d) Determinant $= A^T . A$

e) Inverse $= (A^T . A)^{-1}$

only if $\Delta(A^T . A) \neq 0$

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★ Linear Transformations & Geometry.

2D space → Line
3D space → Plane
More than 3 subjects → Hyperplane
Here, 6 subjects → 6 dimensional space

★ EigenValues & Decomposition.

$$A\vec{v} = \lambda\vec{v}$$

A = covariance matrix
$\vec{v}$ = eigen vector
λ = eigenvalue

⇒ LU Decomposition:

$$A = LU$$

L = Lower triangular
U = Upper triangular

⇒ Singular Value Decomposition (SVD):

$$A = U\Sigma V^T$$

U = Left singular vectors
Σ = Singular values
V^T = right singular vectors